

# Calculus 101

## 1. Standard Derivatives

- $\frac{d}{dx}(x^n) = nx^{n-1}$  (for constant  $n$ )
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(\ln x) = \frac{1}{x}, x > 0$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(k) = 0$  (constant rule)

**Extension for linear inner functions:**

- $\frac{d}{dx}((ax + b)^n) = an(ax + b)^{n-1}$
  - $\frac{d}{dx}(e^{ax+b}) = ae^{ax+b}$
  - $\frac{d}{dx}(\ln(ax + b)) = \frac{a}{ax+b}$
  - $\frac{d}{dx}(\sin(ax + b)) = a \cos(ax + b)$
  - $\frac{d}{dx}(\cos(ax + b)) = -a \sin(ax + b)$
  - $\frac{d}{dx}(\tan(ax + b)) = a \sec^2(ax + b)$
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## 2. Differentiation Rules

- **Sum / Difference:**

$$\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$$

- **Constant Multiple:**

$$\frac{d}{dx}(c f(x)) = c f'(x)$$

- **Product Rule:**

$$\frac{d}{dx}(uv) = u'v + uv' \text{ or } v \frac{du}{dx} + u \frac{dv}{dx}$$

- **Quotient Rule:**

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$$

- **Chain Rule:**

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \text{ (composite functions)}$$


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## 3. Tangents and Normals

- Gradient of tangent at  $(x_1, y_1)$ :  $m_t = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}$

- Equation of tangent:  $y - y_1 = m_t (x - x_1)$
  - Gradient of normal:  $m_n = -\frac{1}{m_t}$  (if  $m_t \neq 0$ )
  - Equation of normal:  $y - y_1 = m_n (x - x_1)$
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## 4. Stationary Points and Nature

- For stationary points, solve  $\frac{dy}{dx} = 0$ .
  - **First Derivative Test:** Check sign of  $\frac{dy}{dx}$  on either side of the point.
  - **Second Derivative Test:**
    - $\frac{d^2y}{dx^2} > 0 \implies$  minimum
    - $\frac{d^2y}{dx^2} < 0 \implies$  maximum
    - $\frac{d^2y}{dx^2} = 0$  (test is inconclusive; use first derivative test; may be a point of inflection)
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## 5. Rates of Change

- If  $y = f(x)$ , then  $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$  (connected rates)
  - For related variables, e.g.  $V = f(r)$  and  $\frac{dr}{dt}$  given, then  $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$ .
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## 6. Integration (Indefinite)

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C$  (for  $n \neq -1$ )
- $\int \frac{1}{x} dx = \ln |x| + C$
- $\int e^x dx = e^x + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$
- $\int \sec^2 x dx = \tan x + C$

### Linear extension:

- $\int (ax + b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$  (for  $n \neq -1$ )
  - $\int \frac{1}{ax+b} dx = \frac{1}{a} \ln |ax + b| + C$
  - $\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$
  - $\int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + C$
  - $\int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + C$
  - $\int \sec^2(ax + b) dx = \frac{1}{a} \tan(ax + b) + C$
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## 7. Definite Integration and Area

- Definite integral:  $\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$
  - Area between a curve  $y = f(x)$  and the  $x$ -axis from  $x = a$  to  $x = b$  (assuming  $f(x) \geq 0$ ):  
Area =  $\int_a^b f(x) dx$
  - If the curve goes below the  $x$ -axis, split into parts using absolute values:  
Area =  $\int_a^b |f(x)| dx$
  - Area between two curves  $y = f(x)$  and  $y = g(x)$  from  $x = a$  to  $x = b$  (where  $f(x) \geq g(x)$ ):  
Area =  $\int_a^b [f(x) - g(x)] dx$
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## 8. Volumes of Revolution (about the $x$ -axis)

- When a region bounded by  $y = f(x)$ , the  $x$ -axis, and lines  $x = a$ ,  $x = b$  is rotated  $360^\circ$  about the  $x$ -axis:  
 $V = \pi \int_a^b [f(x)]^2 dx$
  - Volume of solid formed between two curves  $y = f(x)$  and  $y = g(x)$  rotated about the  $x$ -axis (where  $f(x) \geq g(x) \geq 0$ ):  
 $V = \pi \int_a^b ([f(x)]^2 - [g(x)]^2) dx$
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## 9. Kinematics (Motion in a Straight Line)

Given  $s$  = displacement,  $v$  = velocity,  $a$  = acceleration:

- $v = \frac{ds}{dt}$
  - $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$
  - Integrating:  $v = \int a dt$  (with initial conditions)
  - $s = \int v dt$  (with initial conditions)
  - When initial values  $v(0) = u$ ,  $s(0) = s_0$  are known, use them to find constants of integration.
  - Common form:  $v = \int a dt + C$ , then use  $v(0)$  to find  $C$ .
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## 10. Integration by Substitution (Simple Linear)

- $\int f(ax + b) dx$ , let  $u = ax + b$ , then  $du = a dx$ , so  $dx = \frac{du}{a}$ .  
 $\int f(ax + b) dx = \frac{1}{a} \int f(u) du$
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## 11. Small Increments / Approximate Change (optional, but sometimes seen)

- $\delta y \approx \frac{dy}{dx} \cdot \delta x$  (linear approximation)
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